

Rosen — §6.3: Permutations and Combinations

Selected Exercises

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Exercise 25

One hundred tickets, numbered $1, 2, 3, \dots, 100$, are sold to 100 different people for a drawing. Four different prizes are awarded, including a grand prize (a trip to Tahiti). How many ways are there to award the prizes if

- (a) there are no restrictions?
- (b) the person holding ticket 47 wins the grand prize?
- (c) the person holding ticket 47 wins one of the prizes?
- (d) the person holding ticket 47 does not win a prize?
- (e) the people holding tickets 19 and 47 both win prizes?
- (f) the people holding tickets 19, 47, and 73 all win prizes?
- (g) the people holding tickets 19, 47, 73, and 97 all win prizes?
- (h) none of the people holding tickets 19, 47, 73, and 97 wins a prize?
- (i) the grand prize winner is a person holding ticket 19, 47, 73, or 97?
- (j) the people holding tickets 19 and 47 win prizes, but the people holding tickets 73 and 97 do not win prizes?

Exercise 29

How many 4-permutations of the positive integers not exceeding 100 contain three consecutive integers $k, k + 1, k + 2$, in the correct order

- (a) where these consecutive integers can perhaps be separated by other integers in the permutation?
- (b) where they are in consecutive positions in the permutation?

Exercise 36

How many bit strings contain exactly five 0s and 14 1s if every 0 must be immediately followed by two 1s?